

SANJEEV KUMAR SETHI

NIT Surathkal '26 · Cloud-Native Networking Engineer (Go, eBPF/Cilium, Kubernetes)

sanjeevsethilive@gmail.com | +91-7975965223 | sanjeevsethi.in | linkedin.com/in/Sanjeevliv | github.com/Sanjeevliv

EDUCATION

National Institute of Technology Karnataka
B.Tech. in Electrical & Electronics Engineering

Mangaluru, Karnataka
Dec. 2021 – May 2026

EXPERIENCE

Network Engineering Intern — NITK

Jun. 2025 – Dec. 2025

- Built **WAN aggregation** using **OpenWrt** and **MPTCP** to combine multiple physical links.
- Improved throughput by **60%** while maintaining fault tolerance and session continuity.
- Deployed and tested WAN aggregation on **Raspberry Pi** hardware in a controlled lab environment under real-world traffic simulations.

OPEN SOURCE CONTRIBUTIONS

Cilium (*eBPF-based Networking, Go*)

Feb. 2026

- Authored and Merged [PR #44202](#): Migrated IP handling in the **nodemap** package from **net.IP** to **netip.Addr**, Eliminating heap allocations and ensuring immutable, Type-safe node IP representations.

PROJECTS

Layer 2 Packet Sniffer (*Go, Linux, Raw Sockets*)

Aug. 2026 – Present

- Developed a link-layer packet sniffer in **Go** leveraging **AF_PACKET** raw sockets to capture and inspect live Ethernet frames directly from network interfaces.
- Decoded **Ethernet headers**, extracting source/destination **MAC addresses**, **EtherType**, and encapsulating protocols (**IPv4**, **ARP**) for real-time traffic analysis.

Site Reliability Engineering Platform (*Go, Terraform, Kubernetes*)

Jan. 2026 – Apr. 2026

- Designed and deployed a microservices platform on **GKE Autopilot** using **Terraform**, ensuring reproducible infrastructure and eliminating configuration drift.
- Secured cluster networking with **Network Policies** (controlling **TCP/IP**, **DNS** traffic), private subnets (VPC), and non-root container enforcement.
- Designed **CI/CD** pipelines with **GitHub Actions**, achieving zero-downtime deployments via multi-stage Docker builds and Helm charts.
- Defined **SLIs**, **SLOs**, and **SLAs** targeting 99.9% availability.

TECHNICAL SKILLS

Languages: Go, Python, Bash

Networking: TCP/IP, Raw Sockets (AF_PACKET), MPTCP, DNS, HTTP, Routing, IPv4/IPv6

Operating Systems: Linux, Systemd, Process Management, Shell Scripting

Cloud & DevOps: Kubernetes, Docker, Terraform, Helm, GitHub Actions, GCP

Observability & Reliability: Prometheus, Grafana, SLIs, SLOs, SLAs

CO-CURRICULAR ACTIVITIES

Endurance Sports: Finished 3 Half Marathons (21km each), demonstrating discipline and long-term perseverance.